Syllabus

2023 Fall

Course	Linear Algebra	**Professor**	Askarova Ayjamal
Course No	MSC2020	**Class No**	
Schedule	SAT5	**Grading Eval.**	Relative Evaluation
Other Information			

Profile	Ayjamal Askarova
Course Objectives	1. To become familiar with methods of linear algebra and to acquire the ability to apply it to mathematics, computer science, and information engineering problems 2. To write clearly the concepts of linear algebra such as definition, counterexamples, simple proofs, and to apply the theory to examples. 3. To establish a basic mathematical background for the students who will receive computer or information engineering courses based on linear algebra to acquire the ability to identify, formulate, and solve computer or information engineering problems. 4. To provide the various problem-solving ideas and techniques.
Course Description	This is an undergraduate level course and intended for sophomore year students. The course will cover aspects of vectors in Rn, matrices, linear systems of equations, Gauss-Jordan elimination, determinants, vector spaces, basis, eigenvalues, eigenvectors, LU factorization, singular value decomposition, and QR factorization.
Textbooks	
Other Texts and References	
Class Structure	There are two online lectures per week. There are 13 homework assignments per semester. Every week we conduct a short quiz on previous topics. There is also one optional offline tutorial per week.
Notes	
ABEEK	

Grading

Mid-term	Final exam	Attendance	Assignments	Quiz	Discussion	ETC	Total
20 %	30 %	0 %	20 %	30 %	0 %	0 %	100 %

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Week	Content	Class	Notes
1	**Theme**	Systems of Linear Equations	
	Class Details	Chapters 1.1, 1.2, 1.4 SYSTEMS OF LINEAR EQUATIONS ROW REDUCTION AND ECHELON FORMS THE MATRIX EQUATION Ax = b SOLUTION SETS OF LINEAR SYSTEMS	
	Tests	Homework: p.10 #11,13,15,27 p.22 #13 p.40 #5,7 Quiz #1 (will be uploaded in the e-class)	

2	**Theme**	Linear Transformations	
	Class Details	Chapters 1.7, 1.8, 1.9 LINEAR INDEPENDENCE INTRODUCTION TO LINEAR TRANSFORMATIONS THE MATRIX OF A LINEAR TRANSFORMATION	
	Tests	Homework: p.62 #17,29 p.79 #1,3,5,7,15,17 Quiz #2 (will be uploaded in the e-class)	
3	**Theme**	Matrix Algebra	
	Class Details	Chapters 2.1, 2.2, 2.5 MATRIX OPERATIONS THE INVERSE OF A MATRIX MATRIX FACTORIZATIONS	
	Tests	Homework: p.131 #1,3, 11, 13 (without any engines or CAS) Quiz #3 (will be uploaded in the e-class)	
4	**Theme**	Matrix Algebra	
	Class Details	Chapters 2.7, 2.8, 2.9 APPLICATIONS TO COMPUTER GRAPHICS SUBSPACES OF Rn DIMENSION AND RANK	
	Tests	Homework: p.146 #3,5,7 p.154 #23,25,27 p.160 #9,11 Quiz #4 (will be uploaded in the e-class)	
5	**Theme**	Determinants	
	Class Details	Chapters 3.1, 3.2, 3.3 INTRODUCTION TO DETERMINANTS PROPERTIES OF DETERMINANTS CRAMER'S RULE, VOLUME, AND LINEAR TRANSFORMATIONS	
	Tests	Homework: p.170 #25,27,33,35,37 p.177 #15,24 p.186 #5,7,9 Quiz #5 (will be uploaded in the e-class)	
6	**Theme**	Vector Spaces	
	Class Details	Chapters 4.1 VECTOR SPACES AND SUBSPACES	
	Tests	Homework: p.198 #11,13,17 Quiz #6 (will be uploaded in the e-class)	
7	**Theme**	Midterm examination	
	Class Details	Midterm examination	
	Tests	Midterm examination	
8	**Theme**	Vector Spaces	
	Class Details	Chapters 4.2 NULL SPACES, COLUMN SPACES, AND LINEAR TRANSFORMATIONS	
	Tests	Homework: p.208 ##1-15 (all odd numbers) Quiz #8 (will be uploaded in the e-class)	
9	**Theme**	Eigenvalues and Eigenvectors	
	Class Details	Chapters 5.1, 5.2 EIGENVECTORS AND EIGENVALUES THE CHARACTERISTIC EQUATION	
	Tests	Homework: p.274 #1-19 (all odd numbers) p.282 #3,11,13,15,17 Quiz #9 (will be uploaded in the e-class)	

10	**Theme**	Diagonalization	
	Class Details	Chapters 5.3 MATRIX DIAGONALIZATION	
	Tests	Homework: p.289 #1-19 (all odd numbers) Quiz #10 (will be uploaded in the e-class)	
11	**Theme**	Orthogonality	
	Class Details	Chapters 6.1, 6.2 INNER PRODUCT, LENGTH, AND ORTHOGONALITY ORTHOGONAL SETS	
	Tests	Homework: p.338 #1-25 (all odd numbers) Quiz #11 (will be uploaded in the e-class)	
12	**Theme**	Review (Vectors)	
	Class Details	Chapter 6.1-6.2 PRACTICING	
	Tests	No homework Quiz #12 (will be uploaded in the e-class)	
13	**Theme**	Projections	
	Class Details	Chapters 6.3 ORTHOGONAL PROJECTIONS	
	Tests	Homework: p.354 #1,3,7,11 Quiz #13 (will be uploaded in the e-class)	
14	**Theme**	Orthogonalization	
	Class Details	Chapter 6.4 THE GRAM SCHMIDT PROCESS	
	Tests	Homework: p.360 #1,3,5,7,9 Quiz #14 (will be uploaded in the e-class)	
15	**Theme**	Final examination	
	Class Details	Final examination	
	Tests	Final examination	
16	**Theme**	Make-up classes	
	Class Details	Make-up classes	
	Tests	Make-up classes	